

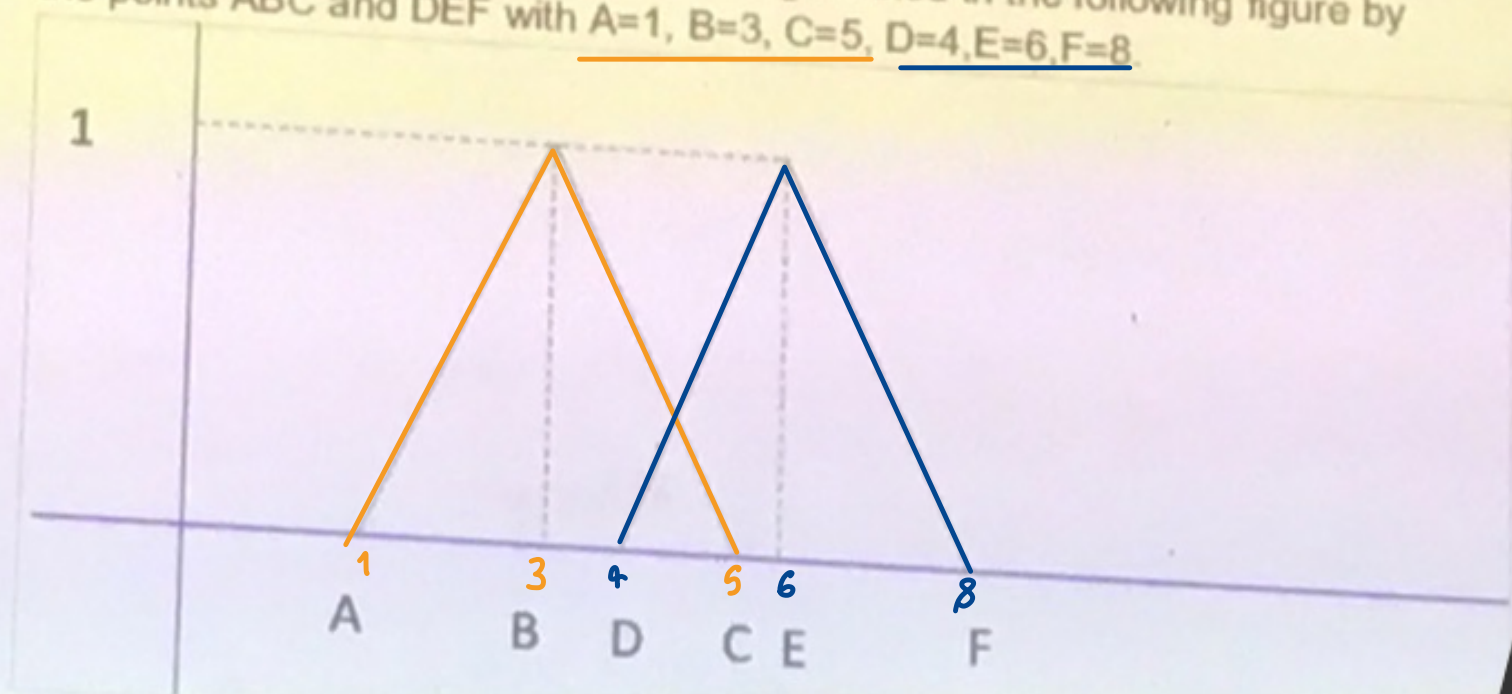
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1. Let us consider the two fuzzy sets f and g defined in the following figure by the points ABC and DEF with $A=1, B=3, C=5, D=4, E=6, F=8$



- Give the equation of the functions f and g
- Compute the intersection and union of f and g by means of t-norm/t-conorm of your choice
- Compute the complement of f and g
- Compute support and core of f and g

CONTINUED
F5

EQUATION of f : **TRIANGULAR**

$$1) \quad f(x) = \begin{cases} \frac{x-1}{3-1} & \text{if } 1 \leq x < 3 \\ \frac{5-x}{5-3} & \text{if } 3 \leq x < 5 \\ 0 & \text{OTHERWISE} \end{cases} \quad g(x) = \begin{cases} \frac{x-4}{6-4} & \text{if } 4 \leq x < 6 \\ \frac{8-x}{8-6} & \text{if } 6 \leq x < 8 \\ 0 & \text{OTHERWISE} \end{cases}$$

2) • INTERSECTION of f AND g

$$(f \wedge g)(x) = \begin{cases} \frac{x-4.5}{4.5-4} & \text{if } 4 \leq x < 4.5 \\ \frac{4.5-x}{5-4.5} & \text{if } 4.5 \leq x < 5 \\ 0 & \text{OTHERWISE} \end{cases}$$

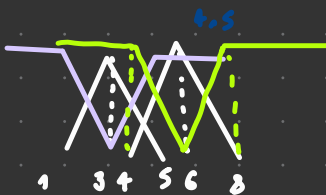
• UNION

$$(f \vee g)(x) = \begin{cases} \frac{x-1}{3-1} & \text{if } 1 \leq x < 4.5 \\ \frac{5-x}{5-3} & \text{if } 3 \leq x < 4.5 \\ \frac{x-4}{6-4} & \text{if } 4.5 \leq x < 6 \\ \frac{8-x}{8-6} & \text{if } 6 \leq x < 8 \\ 0 & \text{OTHERWISE} \end{cases}$$



• COMPLEMENT of f AND g

$$f^c(x) = \begin{cases} \frac{3-x}{3-1} & \text{if } 1 \leq x < 3 \\ \frac{x-3}{5-3} & \text{if } 3 \leq x < 5 \\ 1 & \text{OTHERWISE} \end{cases} \quad g^c(x) = \begin{cases} \frac{6-x}{6-4} & \text{if } 4 \leq x < 6 \\ \frac{x-6}{8-6} & \text{if } 6 \leq x < 8 \\ 1 & \text{OTHERWISE} \end{cases}$$



• SUPPORT $\{x \in X \mid S(x) > 0\}$

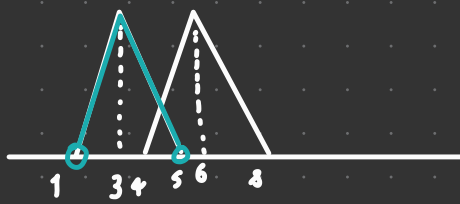
• SUPPORT of $S(x) \Rightarrow (1, 5)$

• SUPPORT of $g(x) \Rightarrow (4, 8)$

• CORE $\{x \in X \mid S(x) = 1\}$

• CORE of $S(x)$ $\{3\}$

• CORE of $g(x)$ $\{6\}$



DISCRETE FS

2. Let us consider the fuzzy set g defined as $g(10)=0.3$, $g(15)=0.5$, $g(20)=1$, $g(25)=1$, $g(30)=0.5$, $g(35)=0.3$, $g(x) = 0$ for all other values of x .

- Compute core and support of g
- Compute the complement of g
- Compute the cardinality g
- Compute the α -cut of g with $\alpha=0.4$

$$g(10) = 0.3$$

$$g(15) = 0.5$$

$$g(20) = 1$$

$$g(25) = 1$$

$$g(30) = 0.5$$

$$g(35) = 0.3$$

$$g(x) = 0 \text{ for all other values}$$

1) CORE of g

$$\{20, 25\}$$

SUPPORT of g

$$\{10, 15, 20, 25, 30, 35\}$$

2) COMPLEMENT of g

$$g^c(10) = 0.7$$

$$g^c(15) = 0.5$$

$$g^c(20) = 0$$

$$g^c(25) = 0$$

$$g^c(30) = 0.5$$

$$g^c(35) = 0.7$$

$$g^c(x) = 1 \text{ for all other values}$$

CARDINALITY of g
(IS THE SUM OF MEMBERSHIP DEGREES)

$$\text{CARD}(g) = (0.3 + 0.5 + 1 + 1 + 0.5 + 0.3) = 3.6$$

α -CUT WITH $\alpha = 0.4$

$$\{15, 20, 25, 30\}$$

ENTROPY

$$\sum (MU(x) \cdot \log(1/MU(x))) =$$

OR USE

$$-\sum (MU(x) \cdot \log(MU(x)))$$

$$H(g) = 0.3 \cdot \log\left(\frac{1}{0.3}\right) + 0.5 \cdot \log\left(\frac{1}{0.5}\right) + 1 \cdot \log\left(\frac{1}{1}\right) + 1 \cdot \log\left(\frac{1}{1}\right) + 0.5 \cdot \log\left(\frac{1}{0.5}\right) + 0.3 \cdot \log\left(\frac{1}{0.3}\right) = 0$$

ENTROPY

$$\sum MU(x) \cdot \log\left(\frac{1}{MU(x)}\right) + (1 - MU(x)) \cdot \log\left(\frac{1}{1 - MU(x)}\right)$$

MEASURE OF FUZZINESS OF μ USING HAMMING DISTANCE

$$F(\mu) = \frac{\sum (|\mu(x) - (1 - \mu(x))|)}{\text{SUPP}(\mu)} \Rightarrow \frac{2\mu(x) - 1}{\text{SUPP}(\mu)}$$

OR DER WEIGHTED AVERAGE WITH $w_i = 1/6$

PICK MEMBERSHIP DEGREE
AND SUM IN ORDERING ORDER

$$1.4/6 + 1.1/6 + 0.5/6 + 0.4/6 + 0.3/6 + 0.3/6$$

THEN MULTIPLY BY CORRESPONDING WEIGHT

YOU CAN SKIP IF THE
WEIGHTS ARE ALL THE SAME

COMBINE OWN OF μ WITH μ DEFINED BY $f(10)=0.7$ $g(15)=0.4$

$$f(20)=1 \quad f(25)=1 \quad f(30)=1 \quad f(35)=0.5 \quad f(40)=0$$

$$w_1 = 0.75 \quad w_2 = 0.25$$

$$\text{ELEMENT 10} \rightarrow f(10)=0.7$$

$$g(10)=0.3 \rightarrow 0.7 \cdot 0.75 + 0.3 \cdot 0.25$$

BIGGEST BETWEEN $f(10)$ AND $g(10)$

$$\text{ELEMENT 15} \quad f(15)=0.7 \quad g(15)=0.5 \rightarrow 0.75 \cdot 0.5 + 0.4 \cdot 0.25$$

↓ SMALLEST BETWEEN ...

$$\text{ELEMENT 25} \quad f(25)=1 \quad g(25)=1 \rightarrow 1 \cdot 0.75 + 1 \cdot 0.25$$

REASONABLE DEFINITION OF $\text{VERY}(\mu)$

$$\mu(x)^2$$

↓ ELEMENT VERY REPRESENTATIVE
OF THE SET

CENTER OF GRAVITY !!! SOLO PER 1 DISCRETE

$$\frac{\sum (\mu(x) \cdot x)}{\sum \mu(x)}$$

↓ WEIGHTED AVERAGE
SUM OF MEMBERSHIP DEGREE

3. Given two propositions P and Q, with truth value $v(P)=0.7$ and $v(Q)=0.4$, compute

- The logical and using min, Lukasiewicz and product t-norm
- The logical or using max, Lukasiewicz and product t-conorm
- $P \rightarrow Q$, using an implication at your choice
- The negation of P

MIN PICK SMALLEST VALUE $\min(P, Q) = 0.4$

$$\text{LUKASIEWICZ}(P, Q) = \max(0, P + Q - 1) = \max(0, 0.7 + 0.4 - 1) = 0.1$$

$$\text{PRODUCT } P \times Q = 0.28$$

$$\text{MAX}(p, q) = 0.7$$

$$\text{Lukasiewicz}(p, q) = \min(1, p+q) = \min(1, 1.1) = 1$$

$$\text{PROJ}_{\text{sum}}(p, q) = p+q - pq = 1.1 - 0.28$$

$p \rightarrow q$

GODEL IMPLICATION IS EASIEST

$$\begin{cases} q & \text{if } p > q \\ 1 & \text{otherwise} \end{cases} \rightarrow \text{GODEL}(p, q) = 0.4$$

LOUCAN IMPLICATION

$$\begin{cases} q/p & p > q \\ 1 & \text{otherwise} \end{cases}$$

NEGATION OF P

$$\text{NEG}(p) = 1 - p = 0.3$$

FUZZY CLUSTERING EXERCISE

!!! NO NEED TO REMEMBER FUZZY CLUSTERING FORMULA

↳ FORMULAS ARE GIVEN, YOU NEED TO KNOW HOW TO APPLY THE FORMULA

$$c_1 = (0, 1)$$

$$c_2 = (1, 0)$$

Wc 2

■ Let $c1 = (0, 1)$, $c2 = (1, 0)$ and $w = 2$. Compute the membership of the points $x1 = (0.75, 0.5)$, $x2 = (0, 0)$ and $x3 = (0.3, 1.1)$ to the clusters whose centroids are $c1$ and $c2$.

Remember that the formula for the degree of membership of object j to cluster i is given by:

$$d_{i,j}^{\frac{2}{1-w}}$$

$$\sum_{i=1}^K \frac{1}{d_{i,j}^{\frac{2}{1-w}}}$$

$K = 1$ K, j

COMPUTE THE EUCLIDEAN DISTANCE

x, y

$$\sqrt{\sum (x(i) - y(i))^2}$$

- TIG
- ANCH
- ALDE
- ANTO
- BESM
- JIMM
- ECI
- PAT
- BAISM
- VTB
- BREKY
- CELE
- MINE
- CURW

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